

Closure by Exhaustion for Same-Scope Operators under Admissibility

Operator saturation, grounded authority, and conservative construction

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Abstract

An operation may construct a new result without acquiring an independent authority to change the qualification of the original object, question, or occurrence. We prove an operator-closure theorem that makes this distinction exact. The original task, result graphs, continuation obligations, and grounds of authorization are specified independently of any proposed realization. A partial operation descends through a complete retained profile exactly when both its domain and its retained output are constant on profile fibres; descended operations compose, and joint constructors obey the corresponding compatibility test. For an original greatest admissible interior, we construct the unique greatest repertoire of additional candidate obligations that preserves that interior. Any set of individually compatible additions is jointly conservative, including under mixed finite execution. A parallel theorem characterizes all conservative additions to a grounded rule system, with arbitrary premise sets, and excludes ungrounded mutual authorization. Faithful changes of representation conjugate the specified operator action, while an exact parameter-erasure test determines whether richer data can answer the unchanged original question. Exhaustion is by witnessed defects: surplus or omitted original results, a domain or output collision, failed local readiness or warrant, an escaping successor, or an unsupported grounded entrance. Named methods such as selection, completion, forcing, model witnesses, and fixed-point constructions are applications of these tests, not premises of their exhaustiveness. Finite assembly, infinitary grounded inference, and limit-result formation receive separate theorems. The resulting closure conserves original admissibility and authority while admitting nontrivial warranted construction, and supplies explicit hypotheses for downstream corpus use.

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1 The operator-closure question

Suppose a construction is claimed to solve an original problem: qualify a carrier, establish an operator application, return a charge or numerical value, preserve a record, or identify a result across presentations. Can additional operator power be obtained by composing methods, moving to a richer representation, introducing a witness, or having several operations authorize one another? The relevant question is what the proposed operation actually establishes about that original problem.

The answer requires three distinctions. Constructing a new output is different from altering the answer about an unchanged original incidence. Deriving an operation from warranted premises is different from supplying its own missing license. Preserving qualification is different from preserving every original observable or identifying all qualified objects. The present paper proves closure results at each of these levels and states the bridges between them.

1.1 Necessity and the original incidence

The Kernel identifies admissibility, standing, reference, and irreversible fixation as necessary functional roles of non-degenerate determinate construction [6]. These roles govern the input, action, output, and record of a mathematical or physical construction. Their governance is not an optional overlay on an otherwise complete physical description. A physical state, carrier, record, or endpoint that is determinate already incurs these roles. Which particular domain condition is met, however, must be established at its original bearer and question. The common necessity of the roles does not identify different questions, erase an input, or supply the outcome of an experiment.

An *original incidence* comprises the bearer and the complete question being answered, with whatever context and occurrence identity the task requires. All warranted presentations of the same determinate incidence agree. A lawful new action can nevertheless create a new incidence. For example, replacing a failed support and running a new test can produce a successful result. It cannot make the retained earlier failed trial a successful trial. This is the operational distinction that a closure theorem must preserve.

The generic constitutive argument belongs to the Kernel paper. Here we prove the additional operator mathematics: what information an action requires, how its authorization is grounded, what happens when actions interact, and exactly when an expanded repertoire preserves the original admissible interior.

1.2 A familiar example: a bridge test and its records

Consider a small model bridge, a specified load, and a test asking whether that bridge carries the load unsupported for a specified duration. A record identifies the bridge, trial, conditions, and result. Four proposals illustrate the problem.

1. A program converts the same record between two file formats. It must preserve the original fields needed to answer the test question. Different bytes need not mean a different test result.
2. A program retains only the word “tested.” Two records with that same retained word may have different outcomes. No function of the word alone can recover both original answers.
3. Someone supports the bridge with a finger. The supported trial may succeed, but its success is not the unsupported trial’s result. A declared change of conditions is a new question; an undeclared change is a false realization of the old one.

4. Two certificates cite each other as the sole reason that the test was passed. Without an independently grounded entrance, their cycle establishes neither certificate. A real test or a sound derivation from independent data can supply such an entrance.

There are also legitimate transformations: a valid calculation may derive a load margin, and a warranted repair may create a bridge that succeeds in a new trial. Their legitimacy is not proved by calling their outputs identical to their inputs. Their domains, results, and warrants are the work to be established. Sections 5, 6, and 10 return to these cases.

1.3 The dependence map

The two principal results are theorems 5.5 and 6.5: a unique greatest conservative operator repertoire and a unique greatest uniformly conservative repertoire of derived rules. The authorization argument starts with the simpler fixed-ground criterion in theorem 6.3; theorem 6.5 then establishes conservation across every ground environment. Theorem 8.2 combines these results with the information and grounding tests into closure of authority for the unchanged original problem.

The proof proceeds through seven steps.

1. Fix the original result graphs and complete question profile independently of the proposed method (Section 2).
2. Prove exact partial-operation descent and parameter erasure, with domain information included (Section 3).
3. Prove sequential and joint assembly, contextual substitution, and the separate bridge from generated uses to original uses (Section 4).
4. Construct the greatest conservative operator repertoire over the original admissible interior (Section 5).
5. Prove the corresponding conservative rule-extension theorem and exclude ungrounded mutual licensing, including infinitary premise sets (Section 6).
6. Prove faithful transport and identify the extra obligations for genuine limit outputs (Section 7).
7. Combine the exact criteria into witnessed exhaustion and an operator-closure theorem, then instantiate the familiar method families (Sections 8–10.5).

1.4 Five tests and three meanings of closure

Three closures must be distinguished. A *generated closure* contains what follows from given operations or rules. A *greatest safe interior* contains states satisfying all original continuation obligations. A *maximal conservative repertoire* contains exactly the candidate additions that preserve a specified original interior. The latter two constructions are related by theorem 5.5; none is identified with the others merely by using the word “closure.”

The recurring tests are the *graph test* for exact realization, the *fibre test* for information loss, the *joint-domain test* for assembly, the *interior test* for conservative reuse, and the *grounding test* for authorization. Every test has a displayed necessary and sufficient criterion or a proved obstruction with explicit hypotheses. An absent proof is an open obligation; it is not itself a counterexample.

The term *bookkeeping* is reserved below for preservation of a stated complete profile. A nonidentity action on that profile may still be a warranted construction. Consequently closure of original

authority is not a claim that all admissible mathematics is the identity function. It says exactly which original answers, licenses, and admissible regions a purported unchanged realization is unable to alter.

1.5 Relation to the foundation stack and literature

The division of labor is explicit. The Kernel supplies necessary functional roles and original-incidence conservation. AMetric supplies pre-fixational discrimination, non-transmission, selector tests, and grounding. Structure of Admissibility constructs the greatest interior. Fixed-Domain Exhaustion develops original-task coverage and witnessed defects. ATS identifies complete operator envelopes and their typed uses. Unus develops faithful interior transport and conservative additions of contracts [6, 7, 8, 9, 10, 11]. This paper proves the operator-specific saturation and authorization results and organizes their family exports. The elementary imported mathematical facts needed by those proofs are stated and proved locally. No theorem here is obtained by citing a downstream application that presupposes this paper.

The mathematical tools have established neighbors. Quotient factorization and compatible operations belong to universal algebra [1]; the relevant additional feature here is preservation of partial domains and independently interpreted original uses. Tarski’s lattice theorem supplies the general fixed-point setting [2]. Transition systems and behavioral comparison provide the setting for continuation profiles [3]; we do not equate nondeterministic trace equivalence with bisimulation. Safety and eventual success have different logical forms [4]. Program reasoning supplies the familiar demand that composition carry its intermediate assertions and domain conditions [5]. Our contribution is the combined exact operator audit, the two maximal conservative extension theorems, and their explicit dependence on original-task interpretation. These results do not replace the existence hypotheses of classical fixed-point, logical, or universal-property theorems.

2 Original tasks, graphs, and same-scope claims

Definition 2.1 (Original answer frame). An original answer frame consists of a set X of identified presentations or states, an index set J , and partial answers $a_j : X \multimap Y_j$. Context, bearer identity, and occurrence identifiers are included among the answers whenever required by the original problem. Introduce a distinct unavailable marker $\dagger_j \notin Y_j$ and totalize each answer as $a_j^\perp : X \rightarrow Y_j \sqcup \{\dagger_j\}$. Its complete profile is

$$P(x) = (a_j^\perp(x))_{j \in J}, \quad Q = P(X), \quad q : X \multimap Q, \quad q(x) = P(x).$$

The frame is fixed independently of the realization being tested. J and X need not be finite, and no algorithm for the complete profile is assumed.

The unavailable marker records the specified partial domain. It is not a third truth value for a well-formed qualification proposition. Lack of evidence that an answer exists is also not evidence that its original domain excludes the input. Whether an implementation covers that domain is a separate question.

Definition 2.2 (Same-scope realization claim). A claim is same-scope when it purports to realize the already specified bearer, original question family, meanings, and claimed uses. Its input interpretation and asserted coverage must be supplied. This describes the claim being assessed, not a guarantee that the claim is correct. A proposal can be same-scope in its claim and fail the graph, information, warrant, or history tests.

This definition does not assume equality with the original answers. It permits erroneous proposals to be evaluated rather than removing them from the comparison class by definition. A scope change must be exhibited in an independently specified coordinate: a new trial, condition, domain, output question, permitted use, or authority premise. Mere disagreement with a preferred answer does not prove that a different task was intended.

Theorem 2.3 (Exact realization and graph exhaustion). *For one original partial answer $a : X \multimap Y$, put $G_a = \{(x, a(x)) : x \in \text{dom } a\}$. Let $T \subseteq X \times Y$ be the interpreted records claimed by a proposed complete realization. Exactly one of the following holds:*

1. $T = G_a$;
2. *there is a surplus record in $T \setminus G_a$ or an omitted record in $G_a \setminus T$.*

The two defect types may coexist. A sound partial realization requires only $T \subseteq G_a$; a missing record defeats a completeness claim, not soundness alone. Two sound complete realizations of the same original answer agree.

Proof. If the sets differ, one of their two containments fails and supplies the indicated member. Conversely either member contradicts equality. Soundness is precisely the first containment. A sound complete realization satisfies both containments and equals G_a , so two such realizations agree by transitivity of equality. $\square$

The same set-difference proof applies to an independently specified relation $G \subseteq X \times Y$ when several outputs are permitted. Soundness is $T \subseteq G$; exact realization of the full relation is $T = G$. A selected single output has the distinct claim that its pair belongs to G , and any claimed canonicity needs its own selector proof. Determinacy of these membership questions does not make every permitted output identical.

Corollary 2.4 (No conservative totalization of an originally unavailable result). *Suppose $x \notin \text{dom } a$. A proposal returning (x, y) as a result of this same original partial answer has a surplus record, for every $y \in Y$. No representation change, additional notation, or later computation makes that record a member of G_a while retaining the same interpreted answer. In contrast, extending an incomplete realization to include a member already in G_a is not excluded.*

Proof. No pair whose first coordinate is x belongs to G_a . The surplus conclusion follows from theorem 2.3. Adding an already original record can preserve the containment defining soundness. $\square$

This is graph rigidity derived from the independently fixed original result relation and exactness of its claimed realization. It does not infer graph rigidity from preservation of the underlying set alone. Nor does it assume that every present implementation is already complete. An operationally required readiness condition will be made explicit in Section 5.

Theorem 2.5 (Minimal complete information). *Let $h : X \multimap H$ be a retained interface. All original answers factor through h if and only if*

$$h(x) = h(y) \implies q(x) = q(y).$$

When they factor, there is a unique $u : H \rightarrow Q$ with $q = uh$. If also $q(x) = q(y)$ implies $h(x) = h(y)$, then u is a bijection. Any two such minimal complete interfaces are uniquely isomorphic by a bijection commuting with their maps from X .

Proof. Factored answers agree on every h -fibre, hence so does the whole tuple. Conversely define $u(h(x)) = q(x)$; the implication gives well-definedness and surjectivity of h gives uniqueness. Each coordinate projection composed with u supplies its reader. The reverse implication makes u injective; it is onto because q is onto. Composing the two bijections with Q gives the comparison, whose uniqueness follows on every image of X . $\square$

If h loses a distinction, the witness is two inputs with the same retained code and different original answers. If h is adequate but not minimal, it retains an unnecessary distinction for this frame. Such surplus coding is not itself an inadmissible computation. If exact reconstruction of the enriched record is an original use, its identity must instead be included in the frame.

Definition 2.6 (Three operator claims). For an operation on X , *qualification preservation* concerns a specified qualification coordinate. *profile preservation* concerns all of P . *construction warrant* concerns the independently specified domain, result, and permitted use of the operation itself. None is substituted for another without a connecting theorem. A profile-preserving operation is called bookkeeping relative to that profile; this permits changes of realization inside a fibre of q .

For instance, sorting a record into a different storage order can preserve its complete semantic profile while changing its byte identity. Computing a new quantity can change the profile and still be warranted by the original construction rules. Merely preserving a success bit proves neither case. The following sections identify the exact algebraic and operational conditions.

3 Exact partial operators and auxiliary data

Let $q : X \rightarrow Q$ be the attained original profile. The same construction applies to different source and target sorts; we start with one carrier to make composition transparent. No admissibility of a particular execution is inferred merely from the existence of a set-theoretic function. The partial-operation criterion is the primitive descent test of Fixed-Domain Exhaustion [9, Theorem 4.3]; its short proof is retained here so that domain preservation and the subsequent erasure test can be inspected together.

Theorem 3.1 (Exact partial-operation descent). *For $F : D \rightarrow X$, $D \subseteq X$, there exists a unique partial operation $\bar{F} : Q \rightarrow Q$ satisfying*

$$x \in D \iff q(x) \in \text{dom } \bar{F}, \quad q(Fx) = \bar{F}(qx) \quad (x \in D) \quad (3.1)$$

if and only if both conditions hold:

$$qx = qy \implies (x \in D \iff y \in D), \quad (3.2)$$

$$qx = qy, \ x, y \in D \implies q(Fx) = q(Fy). \quad (3.3)$$

Failure is witnessed either by equal-profile inputs with different applicability, or by applicable equal-profile inputs with different output profiles.

Proof. The equivalences and commuting equation in (3.1) immediately imply both conditions. Conversely set $\text{dom } \bar{F} = q(D)$ and define $\bar{F}(qx) = q(Fx)$ for $x \in D$. Output congruence makes the value independent of the representative. Domain saturation ensures that every representative of a class in $q(D)$ lies in D , giving the reverse implication for domains. Surjectivity of q forces both the domain and all values of any other exact descended operation. Negating the two conditions gives the displayed witnesses. $\square$

The original operator may contain distinctions finer than a proposed q . When the criterion fails, either the interface must retain the missing distinction or the claimed operation on that interface must change. One cannot repair the collision by simply declaring a preferred representative authoritative. The AMetric fibre test is the general information principle behind this result [7]; here availability is explicitly part of the answer.

Corollary 3.2 (The exact identity condition). *If F descends, it preserves the original profile on its domain exactly when $\bar{F}$ is the identity restricted to $q(D)$. In particular, descent alone does not imply identity. All total profile-preserving maps $N : X \rightarrow X$ are precisely the families of maps $N_z : q^{-1}(z) \rightarrow q^{-1}(z)$ assembled over $z \in Q$.*

Proof. The equation $q(Fx) = qx$ is equivalent to $\bar{F}(z) = z$ for every attained domain value. For the second statement, $qN = q$ places each image in its original fibre. Restriction gives the maps N_z , and assembling such maps gives $qN = q$. $\square$

These maps form a monoid under composition; their invertible members form the product of the permutation groups of the fibres. This is a classification of semantic preservation, not a claim that every such map is computable or physically available. A map that changes the profile can also be warranted as a different original action. The rule that computes a successor, for example, need not redescribe its input number as itself.

Theorem 3.3 (Exact removal of auxiliary data). *Let $e : Z \rightarrow X$ forget an auxiliary coordinate or richer representation. Let $r : Z \rightarrow Y \sqcup \{\dagger\}$ be the totalized result being claimed, including availability. There is a unique $d : e(Z) \rightarrow Y \sqcup \{\dagger\}$ with $r = de$ if and only if*

$$e(z) = e(z') \implies r(z) = r(z'). \quad (3.4)$$

An answer on all original X additionally requires coverage $e(Z) = X$ or separately warranted answers on the uncovered inputs. When (3.4) fails, the witnessing pair prevents every decoder using only $e(z)$ from realizing both required results.

Proof. A factored function is constant on each fibre. Conversely define $d(x)$ as the common r -value of preimages of $x \in e(Z)$. Constancy makes this well-defined and existence of preimages gives uniqueness on the image. At a collision, a sole-input function would have to take two different values at the same argument, which is impossible. No value outside the image is determined by this construction. $\square$

Taking Z to be the permitted pairs (x, α) yields the exact test for eliminating a selector, oracle answer, measure parameter, completion witness, normalization choice, or returned foreign-state certificate. The set of permitted pairs may depend on x ; it need not be a product. It must, however, be the original family of relevant possibilities, not a family pruned to a preferred output. The result r may itself be an entire original answer profile.

There are three separate obligations. Constancy proves that the information suffices. An original graph argument proves that the common answer is correct. An execution argument proves that the proposed method uses licensed operations. A constant wrong answer passes constancy and fails correctness; an unnecessarily illicit intermediate computation can return a correct constant answer and fail the execution contract.

Corollary 3.4 (No independent additional original answer). *Let q be complete for the original family and let $b : X \rightarrow Y \sqcup \{\dagger\}$ be another required original answer. Then b is determined by q exactly when it is constant on q -fibres. A pair with equal q and unequal b refutes completeness of q for that enlarged original family. Conversely, if every original answer factors through a proposed coordinate v , adding more presentations cannot create an independent original answer at fixed v .*

Proof. Apply theorem 3.3 to the surjection q , or to the attained image of v . Completeness means factorization of every required coordinate, so a nonfactoring coordinate contradicts it. $\square$

This is the bridge required by a downstream “no additional register” argument. Completeness of v must be established from the domain’s original observables. Several mathematically independent coordinates may all be required in one complete profile; neither qualification nor the name “one kernel” limits their number. A proposed rank, dimension, or singleton result requires the corresponding independent separation and existence argument. Section 10.5 makes the relevant export explicit.

3.1 Selectors and equivariance

If a group G acts on an input space and candidate outputs, and the eligible candidate sets satisfy $E_{gx} = gE_x$, an equivariant selector must choose an element fixed by the stabilizer G_x of its input. Indeed, for $g \in G_x$, equivariance gives $g s(x) = s(gx) = s(x)$. Conversely, supplied eligible stabilizer-fixed choices at one representative of each orbit extend consistently by $s(gx) = g s(x)$; equal representations of an orbit point differ by a stabilizer element. Orbitwise choices are supplied data, not a conclusion of the argument. This is the exact selector criterion developed in AMetric [7].

The criterion prohibits a claimed invariant selection when the eligible set has no stabilizer-fixed member. It allows a selector when the requisite structure is present. Invariance of a selected result does not identify that result with its input. If the selector affects the original result after its auxiliary information is forgotten, theorem 3.3 provides the additional no-transfer test.

4 Composition, joint construction, and complete uses

Theorem 4.1 (Exact composition of partial operators). *Suppose $F, G : X \rightarrow X$ satisfy theorem 3.1. Then $F \circ G$ also satisfies it, and*

$$\overline{F \circ G} = \bar{F} \circ \bar{G}$$

as partial maps, including equality of domains. Every finite composite therefore descends. If each factor preserves the complete profile, so does the composite.

Proof. For equal-profile inputs x, y , domain saturation for G gives equal availability. If available, its output congruence gives $q(Gx) = q(Gy)$. Domain saturation for F then gives equal availability of the second step, and its congruence gives equal final profiles. This proves the two descent conditions. Moreover

$$x \in \text{dom}(FG) \iff qx \in \text{dom } \bar{G} \text{ and } \bar{G}(qx) \in \text{dom } \bar{F}.$$

The commuting equations identify the result with $\bar{F}\bar{G}(qx)$. Induction proves the finite claim. Composing restricted identity actions preserves the profile wherever the composite is defined. $\square$

An operation can be newly expressible in a language without being a new source of original authority. For example, a signature containing one step can acquire a useful two-step abbreviation. The abbreviation is not necessarily that one step or identity; its action is the composite already justified by the primitives. Conversely, a candidate outside the generated language is not thereby impossible. It must satisfy its own original graph, interface, and warrant obligations, or be covered by an independently proved generation theorem.

Theorem 4.2 (Exact joint-domain descent). *For $i = 0, \dots, n$, let $q_i : X_i \rightarrow Q_i$, and let $C : D \rightarrow X_0$ be an original constructor with $D \subseteq \prod_{i=1}^n X_i$. Put $H = \prod_i q_i$. There is a unique exact partial constructor $\bar{C} : \prod_i Q_i \rightarrow Q_0$ precisely when*

$$\begin{aligned} H(x) = H(y) &\implies (x \in D \iff y \in D), \\ H(x) = H(y), \ x, y \in D &\implies q_0(Cx) = q_0(Cy). \end{aligned}$$

Every failure has a joint applicability or retained-output collision. For finite n the product map H is onto. More generally, the result holds on $H(\prod_i X_i)$ without any choice assumption for an infinite product.

Proof. Apply theorem 3.3 to the totalized output $(q_0 C)^\perp$ and the map H . Availability constancy is its first clause, and value constancy its second. The induced domain is $H(D)$. Surjectivity in the finite case follows by choosing preimages successively for the finitely many coordinates. The factorization on the attained image requires no such extension. $\square$

Separate qualification of all arguments does not prove the first clause. If the original constructor accepts equal pairs $(0, 0)$ and $(1, 1)$ but rejects mixed pairs, replacing each argument by a constant “qualified” code loses the compatibility test. This is a witnessed failure even though no argument has a third standing value.

Theorem 4.3 (Finite assembly and first semantic defect). *Consider a many-sorted language of finite expression trees with variables, fixed constants, and strict partial constructors. Strictness means that an unavailable child makes the parent unavailable. Suppose each constructor satisfies theorem 4.2. Then the totalized output profile of every expression is determined by the retained profiles of its leaves. If two equal-profile leaf assignments give different totalized answers, some constructor instance violates its joint-domain or output condition, unless a purportedly fixed constant or leaf interpretation differs.*

Proof. Induct on the tree. Leaves give their specified values. For a constructor, the induction hypothesis gives equal availability and profiles of every child. A missing child makes both outputs unavailable by strictness. Otherwise the joint criterion gives equal applicability and, if applicable, equal retained outputs. For a discrepancy, descend to a discrepant child whenever one exists. If none does, the present constructor violates its criterion. Finite descent reaches either that constructor or an inconsistent leaf or constant. $\square$

This witness concerns a semantic discrepancy between two executions. It does not say that the first illegal action of a single execution must affect its final output. Legality of each executed step is a separate trace condition. For a language with conditionals or non-strict constructs, use its actual evaluation rule, including guards and the set of evaluated children. A branch condition must itself descend; importing a branch oracle is not licensed by a theorem about strict trees.

Definition 4.4 (Complete generated-use profile). Fix a finite-expression language with original interpretations and an original observation family on its output sorts. A use is an interpreted expression followed by a specified observation, with all unavailable cases recorded. The generated-use profile $P_{\mathcal{L}}(x)$ is the tuple of answers to every such use on input x . Define $x \equiv_{\mathcal{L}} y$ when these tuples agree.

The set of uses can be infinite even though each expression is finite. Every required comparison context belongs to the family; selected favorable tests do not give its complete profile. For deterministic primitive continuations, closure of uses under prefixing implies that $\equiv_{\mathcal{L}}$ is a congruence for those continuations, including their domains, provided availability is observable. This is the finite-use envelope setting of ATS and Fixed-Domain Exhaustion [10, 9].

Theorem 4.5 (Contextual substitution and original coverage). *Suppose the language contains the relevant compositions and contexts. Two implementations with equal answers in every declared context are interchangeable in those contexts. If every independently original answer $a_j^\perp$ is proved equal to an interpreted generated use followed by a retained-output reader, equality of generated-use profiles implies equality of complete original profiles. If, conversely, every generated use factors through q , the two equivalence relations coincide.*

Proof. The first assertion is equality of the specified contextual answers. For the second, equal generated profiles give equal values of the interpreted use for each j , hence equal values after its reader. The independently established interpretation identifies these with $a_j^\perp$. Thus all coordinates of P agree. Conversely, factorization of each generated use through q gives equality of all generated answers whenever q agrees. Combining the implications gives equality of equivalence relations. $\square$

The interpretation equality is substantive. An enumeration of operation names, a successful sample computation, or a finite registry does not establish it. On the other hand, the semantic results of Sections 2 and 3 apply to arbitrary complete original query families without a finite generation hypothesis. The coverage theorem is the bridge needed when one wishes to discharge those semantic obligations by a specific implementation language.

4.1 History-sensitive and endpoint equivalence

Let $\mathcal{H}$ be a set of execution histories with endpoint map $v : \mathcal{H} \rightarrow X$. If the original question is only an endpoint property a , its answer is av . If it also asks whether every executed action was licensed or whether an earlier trial failed, those coordinates belong to its history profile. An endpoint interface is complete for that larger task exactly when those history answers are constant on its fibres, by theorem 3.3.

Proposition 4.6 (Retention without endpoint absorption). *Suppose a history extension retains the identity and result of every earlier occurrence. If an earlier occurrence failed its original qualification, every such extension still contains that failed occurrence. A later successful endpoint does not erase it. This does not imply that all later endpoints fail.*

Proof. The retained occurrence has the same bearer, question, and original result. By original-answer conservation its result remains the same. The proposition “some retained occurrence failed” therefore remains true. No premise identifies a later endpoint question with that earlier occurrence’s question, so no negative conclusion about the later endpoint follows. $\square$

For the bridge example, the file conversion must preserve the old trial’s result. A repaired bridge may pass a new trial. A program that reads and discards an irrelevant random bit may have the same endpoint function as a deterministic program, but the histories differ if the original task records that read. If the read was forbidden, its execution remains forbidden; this does not invalidate every alternative implementation of the same endpoint function.

5 The unique greatest conservative operator repertoire

We now pass from information preservation to warranted reuse. The original contract is specified before any extension. Its states may be mathematical descriptions, physical configurations, or records of construction. The following is the safety form of the admissible interior developed in Structure of Admissibility and Unus [8, 11]. We give the short construction because the operator-saturation theorem depends on its exact quantifiers.

Definition 5.1 (Original reuse contract). Fix X , a local qualification set $A \subseteq X$, a set R of labeled proposed transitions, and a subset $W \subseteq R$ of transitions with their original warrants. Any required availability or readiness belongs to A . Let

$$B = \{x \in A : \text{every } (x, \ell, y) \in R \text{ lies in } W\}, \quad \text{Succ}(x) = \{y : \exists \ell (x, \ell, y) \in R\},$$

and set $\Phi(U) = \{x \in B : \text{Succ}(x) \subseteq U\}$ for $U \subseteq X$. A region U is sound if $U \subseteq \Phi(U)$.

All declared possible outcomes are checked. A required operation with no result cannot become safe by omitting its edge: its readiness condition fails in A . An optional operation outside its domain imposes no edge at that input. These are different original contracts, and their choice cannot be deferred until a proposed implementation fails.

Lemma 5.2 (Greatest interior and finite unsafe paths). *The set*

$$S = \bigcup \{U \subseteq X : U \subseteq \Phi(U)\}$$

is the unique greatest sound region and satisfies $S = \Phi(S)$. Moreover $x \in S$ exactly when every finite R -path from x , including the path of length zero, ends in B . Hence $x \notin S$ has a finite path ending outside B . No finite-carrier or finite-branching hypothesis is required.

Proof. If x lies in a sound U , then $x \in B$ and $\text{Succ}(x) \subseteq U \subseteq S$, giving $S \subseteq \Phi(S)$. Monotonicity makes $\Phi(S)$ itself postfix, so it is included in the union S ; this proves fixedness. Greatestness is immediate from the union. Every path from a sound set remains in that set by induction, so its endpoint lies in B . Conversely the set of states whose finite paths end in B lies in B by length zero, and prepending an edge shows it is successor-closed. It is sound, hence contained in S . Negating the universal path condition gives a finite bad path. $\square$

This statement concerns safety. Eventual production of a desired result is not the same universal finite-path condition. Nonemptiness of S also requires a witness: a nonempty sound region suffices, while the abstract greatestness theorem allows $S = \emptyset$.

Definition 5.3 (Candidate operator obligations). Fix a set $\mathcal{E}$ of independently interpreted candidate packages in the declared original use envelope. For $e \in \mathcal{E}$, specify a local obligation set A_e , transition relation T_e , and warranted subrelation $W_e \subseteq T_e$. Set

$$B_e = \{x \in A_e : T_e[x] \subseteq W_e[x]\}, \quad \Psi_e(U) = \{x \in B_e : T_e[x] \subseteq U\}.$$

For $E \subseteq \mathcal{E}$, retain the old contract and add these obligations:

$$\Phi_E(U) = \Phi(U) \cap \bigcap_{e \in E} \Psi_e(U), \quad S_E = \nu \Phi_E.$$

Here $\nu \Phi_E$ is constructed by lemma 5.2; labels can be tagged by e so their warrants remain distinguishable. An empty intersection is X .

The candidate universe is fixed independently of which packages pass the test. It can contain proposals with false warrants, failed readiness, or unsafe outputs. A claim of maximality relative to all original candidates requires that universe to cover the original use envelope. A physically executable candidate additionally needs the domain's realization evidence; membership in a set of functions does not provide an apparatus.

For a deterministic partial candidate $f_e : D_e \rightarrow X$ and a set K_e of inputs requiring availability, the readiness part of A_e is $(X \setminus K_e) \cup D_e$. Then $x \in K_e \setminus D_e$ witnesses failed readiness. Additional resource or applicability conditions may also be conjuncts of A_e . When the complete profile is claimed unchanged, the independent information conditions of theorem 3.1 and corollary 3.2 must be included as further tests; preservation of S alone does not imply them.

Theorem 5.4 (Exact conservative extension). *For every $E \subseteq \mathcal{E}$, $S_E \subseteq S$. Equality holds exactly when, for every $e \in E$ and every $x \in S$,*

$$x \in A_e, \quad T_e[x] \subseteq W_e[x], \quad T_e[x] \subseteq S. \quad (5.1)$$

Equivalently, $S \subseteq \Psi_e(S)$ for every added package.

Proof. Any Φ_E -sound set is Φ -sound, so $S_E \subseteq S$. The displayed conditions make S sound for every Ψ_e and therefore for Φ_E , since $S = \Phi(S)$. Greatestness then gives $S \subseteq S_E$. Conversely $S = S_E = \Phi_E(S_E)$ implies all the conditions. Expanding B_e gives their stated form. $\square$

Theorem 5.5 (Maximal conservative operator repertoire). *Define*

$$\mathcal{E}_S = \{e \in \mathcal{E} : S \subseteq \Psi_e(S)\}.$$

Then

$$S_E = S \iff E \subseteq \mathcal{E}_S. \quad (5.2)$$

Thus $\mathcal{E}_S$ is the unique greatest repertoire of candidates preserving the original interior. Arbitrary unions of preserving repertoires preserve it. Every repertoire containing a candidate outside $\mathcal{E}_S$ has interior strictly smaller than S . These statements allow arbitrary set-indexed repertoires and arbitrary branching.

Proof. By theorem 5.4, preservation is exactly the conjunction of the individual membership conditions defining $\mathcal{E}_S$. Therefore preserving repertoires are precisely its subsets, which proves greatestness and arbitrary-union closure. An incompatible member prevents equality, and the general containment $S_E \subseteq S$ then gives strictness. $\square$

The theorem derives maximality from an independent local criterion; it does not assume that all admitted operations are identity, already named primitives, or part of a finite registry. In particular, it distinguishes new useful notation, new derived actions, and proposed additional obligations. Their mathematical novelty is compatible with conservation of the original interior when the criterion holds. If S is empty, all the conditions are vacuous, so no existence claim can be exported from maximality alone.

Theorem 5.6 (Exact defect witnesses for repertoire closure). *A candidate e is outside $\mathcal{E}_S$ if and only if there is an $x \in S$ with at least one of the following:*

1. $x \notin A_e$;
2. a proposed transition from x that is absent from W_e ;
3. a proposed transition $x \rightarrow y$ followed by a finite original R -path from y to a point outside B .

The last path may have length zero. Each witness defeats preservation of the original complete interior by any repertoire containing e .

Proof. Negating (5.1) gives a failed local condition, an unwarranted transition, or a successor outside S . The latter is equivalent by lemma 5.2 to the indicated finite suffix. Conversely each witness violates one condition, so theorem 5.5 applies. $\square$

This is the exhaustive operator classification. The alternatives may overlap. They are not an algorithm for deciding an infinite contract or finding its witness. If a warrant is merely unknown, the mathematical assertion that it is absent still requires proof.

Corollary 5.7 (Mixed finite execution and derived operators). *Every finite execution starting in S and using original transitions and transitions from any $E \subseteq \mathcal{E}_S$ remains in S , and every executed transition has its required warrant. A derived partial operation obtained by such a finite execution has outputs in S on its actual domain. It belongs to $\mathcal{E}_S$ if its declared readiness holds throughout S and a sound composition rule supplies the warrant for its derived use.*

Proof. Original edges from S remain in S and are warranted by fixedness. Added edges do so by (5.1). Induct on execution length, without requiring successive edges to have the same label or origin. The endpoint condition follows. Together with the additional readiness and derived-use warrant hypotheses, it is exactly the extension criterion for the composite package. $\square$

Partial safety is not totality: if f and g preserve S wherever defined, $g \circ f$ is defined only when f is available and its result lies in $\text{dom } g$. A demanded composite use must establish both. The theorem also tests all additions against the same S . Individually safe systems with different interiors can fail under interleaving; Section 10 gives the three-state witness.

For many-input constructors, independently specified partner contexts can be included in the labeled use relation. Alternatively, prove directly that every invoked tuple of safe arguments satisfies the joint domain and warrant and produces a safe output. Structural induction then gives term safety. Testing only equal or specially chosen partner tuples does not prove arbitrary compatibility, and allowing the tested region itself to choose its permitted partners can destroy the union property used in lemma 5.2.

6 Grounded authority and maximal conservative rules

Operator safety and grounds of authorization are related but distinct. A transition's warrant must come from the original domain conditions, not from the fact that a proposed procedure would be useful if licensed. This section makes that dependence exact for arbitrary rule systems. It extends the AMetric no-bootstrap result [7] with conservative macro saturation and a grounded-entrance theorem. Theorem 6.3 supplies the fixed-ground criterion; theorem 6.5 supplies the uniform result needed when a derived rule is intended for reuse across ground environments.

Definition 6.1 (Grounded rule system). Let $\mathcal{J}$ be a set of judgments, $B_0 \subseteq \mathcal{J}$ independently grounded judgments, and $\mathcal{R}$ a set of rules (A, j) , with arbitrary premise set $A \subseteq \mathcal{J}$ and conclusion $j \in \mathcal{J}$. A set T is rule-closed when $A \subseteq T$ and $(A, j) \in \mathcal{R}$ imply $j \in T$. Put

$$C = \text{Cl}_{\mathcal{R}}(B_0) = \bigcap \{T \subseteq \mathcal{J} : B_0 \subseteq T \text{ and } T \text{ is rule-closed}\}.$$

The family contains $\mathcal{J}$. Intersections of rule-closed sets are rule-closed, so C is the least rule-closed set containing B_0 .

Judgments may include original domain incidences, source identifications, constructor warrants, and authorization of an interpreted transfer. Infinite premises and infinite registries are permitted. Their original meaning and soundness must be justified; choosing a rule set does not prove that its conclusions are true.

Lemma 6.2 (Semantic soundness and its completeness boundary). *Let $H \subseteq \mathcal{J}$ be the semantically warranted original judgments. If $B_0 \subseteq H$ and every rule is sound in the sense that $A \subseteq H$ implies $j \in H$, then $C \subseteq H$. Equality $C = H$ requires the additional assertion that every judgment in H is grounded by this system.*

Proof. H is a rule-closed superset of B_0 , hence contains the defining intersection. The converse containment does not follow from this reasoning and is precisely the stated additional assertion. $\square$

The closure operation is extensive, monotone in its grounds, and idempotent. Extensivity and idempotence follow from least closedness. For monotonicity, every closed superset of a larger ground set contains the least closed superset of a smaller one. We will use these facts without any finitary assumption.

Theorem 6.3 (Exact conservative rule extension at fixed grounds). *Let M be a candidate set of added rules and $F \subseteq M$. Then*

$$\text{Cl}_{\mathcal{R} \cup F}(B_0) = C \iff \text{for every } (A, j) \in F, A \subseteq C \implies j \in C.$$

Consequently the unique greatest base-relative conservative rule repertoire is

$$M_C = \{(A, j) \in M : A \subseteq C \implies j \in C\}.$$

All its subfamilies, and arbitrary unions of such subfamilies, preserve C . A nonconservative added rule has a witness $A \subseteq C$ with $j \notin C$.

Proof. Adding rules can only enlarge the least closure. If the condition holds, C is closed under the old and new rules, giving the opposite containment by minimality. Conversely equality makes C closed under the added rules. The remaining conclusions are the pointwise restatement of this equivalence. $\square$

Corollary 6.4 (Exact conservation under added grounds and rules). *Let $D_0 \subseteq \mathcal{J}$ be proposed additional grounds and F additional rules. Then*

$$\text{Cl}_{\mathcal{R} \cup F}(B_0 \cup D_0) = C$$

if and only if $D_0 \subseteq C$ and C is F -closed. For conservation uniformly over every B_0 , the conditions are $D_0 \subseteq \text{Cl}_{\mathcal{R}}(\emptyset)$ and the uniform macro criterion of theorem 6.5.

Proof. Equality forces every added ground into C and makes C closed under all added rules. Conversely those conditions make C a closed superset of the enlarged grounds, while monotonicity gives the other containment. Uniform conservation at empty grounds forces the displayed condition on D_0 ; conservation at each macro's premises forces its derivability. Conversely those hypotheses place D_0 in every original closure and make every original closed set closed under F , proving uniform conservation. $\square$

An oracle certificate, extra measurement, or proposed axiom must therefore be assessed as an actual new ground when that is its role. If it supplies a semantically warranted fact outside an incomplete former closure, it can legitimately extend knowledge. It cannot simultaneously be claimed to leave that former grounded closure unchanged. Nor does addition as a premise prove the premise's original truth.

A rule with currently ungrounded premises can satisfy this fixed-base criterion vacuously. It need not be a sound rule for a different environment. The following stronger theorem removes that loophole when a macro is intended for unrestricted reuse.

Theorem 6.5 (Uniformly conservative macro saturation). *For $F \subseteq M$, the following are equivalent:*

- (i) *For every $B \subseteq \mathcal{J}$, $\text{Cl}_{\mathcal{R} \cup F}(B) = \text{Cl}_{\mathcal{R}}(B)$.*
- (ii) *For every $(A, j) \in F$, $j \in \text{Cl}_{\mathcal{R}}(A)$.*
- (iii) *Every $\mathcal{R}$ -closed set is F -closed.*

Thus

$$M_{\text{unif}} = \{(A, j) \in M : j \in \text{Cl}_{\mathcal{R}}(A)\}$$

is the unique greatest uniformly conservative macro repertoire. Every set of such macros can be added simultaneously without changing any original grounded closure. A failed macro has its own premise set as a witnessing ground environment. No bound on arity, grounds, or rule count is required.

Proof. For (i) $\Rightarrow$ (ii), choose $B = A$. The added rule puts j in the extended closure, so equality puts it in the original one. For (ii) $\Rightarrow$ (iii), if $A \subseteq T$ and T is $\mathcal{R}$ -closed, minimality gives $\text{Cl}_{\mathcal{R}}(A) \subseteq T$, hence $j \in T$. For (iii) $\Rightarrow$ (i), the original closure of B is already closed under F , so it contains the extended closure; adding rules gives the reverse containment. Greatestness and arbitrary-union closure follow because the condition tests each macro separately. If (ii) fails at (A, j) , the extended closure of A contains j whereas the original one does not. $\square$

This theorem is a rigorous no-emergence result for derived rules. Combining already derived rules, including infinitely many of them, cannot establish a judgment outside the original consequence closure. A newly discovered proof can establish that a judgment already belongs to that closure; it need not enlarge the semantic consequence relation. Computational convenience, a shorter derivation, and a new independent authority are different claims.

For a specified family of admissible ground environments, an intermediate criterion is available: test every added rule on every original closure from that family. The greatest compatible macro set is the intersection of the corresponding M_C . Universal conservativity should not be inferred from a single environment with unreachable premises.

Theorem 6.6 (No ungrounded mutual licensing). *Let $U \subseteq \mathcal{J}$ be a family of proposed licenses. If $B_0 \cap U = \emptyset$ and every rule with conclusion in U has a premise in U , then*

$$\text{Cl}_{\mathcal{R}}(B_0) \cap U = \emptyset.$$

Proof. The complement $\mathcal{J} \setminus U$ contains B_0 . It is rule-closed: a rule with all premises outside U cannot conclude in U by the hypothesis. The least closure therefore remains in the complement. $\square$

Theorem 6.7 (An actual grounded entrance is necessary). *If $C \cap U \neq \emptyset$, then either $B_0 \cap U \neq \emptyset$ or there exists a rule $(A, j) \in \mathcal{R}$ with*

$$j \in U, \quad A \subseteq C \setminus U.$$

Thus a family reached from the grounds has an entrance whose premises are actually grounded outside that family, not merely an abstract possible entrance with unavailable premises.

Proof. Suppose neither alternative holds. Then $C \setminus U$ contains B_0 . A rule applicable to this set has all its premises in C , so its conclusion belongs to C ; absence of the displayed entrance places that conclusion outside U . Thus $C \setminus U$ is rule-closed. Minimality would give $C \subseteq C \setminus U$, contradicting $C \cap U \neq \emptyset$. $\square$

The existence of one entrance does not prove every license in U . Nor does failure of theorem 6.6's hypothesis prove reachability. The entrance theorem gives a necessary condition with genuinely grounded premises; a concrete derivation and soundness still establish the desired result.

6.1 Infinite premises and the distinction from circular consistency

Take judgments $p_0, p_1, \dots, q$, grounds $\{p_0\}$, rules $p_n \Rightarrow p_{n+1}$, and the rule $\{p_n : n \in \mathbb{N}\} \Rightarrow q$. The grounded closure contains all p_n and then q . No finite stage of applying the successor rules and this final rule reaches q . The theorems above cover this system, while a claim that every consequence has a finite certificate would require a separate compactness or finitary result.

In contrast, for the singleton circular rule $q \Rightarrow q$ with empty grounds, the least grounded closure is empty. The set $\{q\}$ is a larger closed set, but choosing it does not supply the missing warrant. Coinductive definitions, cyclic proof systems with proved global soundness conditions, and independently solved implicit equations can be legitimate. Their additional semantics and soundness are the required entrance; they are not obtained by repeating the circle.

In the bridge example, two mutually citing certificates with no original test or justified inference are the second case. A derivation from independently measured dimensions and a justified load model can be a grounded entrance. Whether that model warrants the specified physical trial remains a domain question rather than a consequence of the rule syntax.

7 Faithful transport, iteration, and limit outputs

7.1 What a change of representation preserves

Suppose two presentation spaces X_1, X_2 have complete original-profile maps $q_i : X_i \rightarrow Q$ for the same original questions. A translation $T : X_1 \rightarrow X_2$ is faithful for that frame when $q_2 T = q_1$. The equality is a property to prove from the interpretations, not a consequence of using the word “translation.”

Proposition 7.1 (Forced coincidence at the complete interface). *If $T, T' : X_1 \rightarrow X_2$ both preserve the same complete original frame, then $q_2 T = q_2 T'$. If a normalization $n : X_2 \rightarrow N$ identifies exactly the equal- q_2 presentations, then $nT = nT'$. Pointwise equality of T, T' follows if q_2 is injective. Preservation of one qualification coordinate alone does not supply these conclusions about the other original coordinates.*

Proof. Both composites with q_2 equal q_1 . Equal q_2 -values imply equal normalization values by the stated kernel condition. If q_2 is injective, equality after q_2 implies equality before it. $\square$

Theorem 7.2 (Transport of operator actions). *Let $h : X \rightarrow Y$ be a bijection with independently faithful original interpretations. For a partial action $G : Y \multimap Y$, define $F = h^{-1}Gh$ on $h^{-1}(\text{dom } G)$. Then $hF = Gh$ on that domain, finite compositions are transported by the same conjugation, and*

$$h(\text{Fix } F) = \text{Fix } G.$$

If local qualification and transitions are transported by h , so that $B_Y = h(B_X)$ and $\text{Succ}_Y(hx) = h[\text{Succ}_X(x)]$, then $S_Y = h[S_X]$. Warrant preservation additionally requires the corresponding tagged transitions to have matching warrant status.

Proof. The action equation follows by cancellation of hh^{-1} , with domains matched by the definition. The same cancellation proves composition, including intermediate definedness. $F(x) = x$ is equivalent to $G(hx) = hx$, proving the fixed-point equation. A set $U \subseteq X$ is locally qualified and successor-closed exactly when $h[U]$ has those properties in Y . Transport the greatest sound regions in both directions. Warrant transport is a distinct coordinate of the stipulated transition interpretation. $\square$

This theorem does not construct a fixed point. It transports one when it exists, and transports the hypotheses of an independently applicable existence theorem when the relevant structure is faithfully included. Banach's contraction, Brouwer's topological, or Tarski's order hypotheses are not supplied by qualification preservation. Similarly, a universal property's existence and mediating-arrow equations must be proved in the stated category before a faithful equivalence can transport them.

For a noninjective quotient there need be no unique lifted action on X . The exact quotient action is instead supplied by theorem 3.1. Distinct realizations in a fibre may realize that same action. A return map can select a realization only under its own domain and warrant conditions. Even identity forward and return translations leave an intervening nonidentity operation nonidentity. What is conserved is its independently identified original action and qualification.

7.2 Iteration within the original interior

Finite iteration is covered by theorem 4.1 and corollary 5.7. For an existing infinite execution $x_0, x_1, \dots$ whose each step is one of the compatible transitions and with $x_0 \in S$, every x_n belongs to S , by induction on n . Existence of such an infinite execution is a separate assertion, and this argument says nothing about an additional endpoint not among its finite stages.

Theorem 7.3 (Exact limit-constructor test). *Let $\mathcal{T}$ be the independently specified space of traces or approximation families relevant to a limit claim, with retained trace interface $e : \mathcal{T} \rightarrow H$. Let $L : D \rightarrow X$, $D \subseteq \mathcal{T}$, be its proposed endpoint constructor. Then:*

1. *A unique exact endpoint-profile operation on $e(\mathcal{T})$ exists if and only if the totalized map $(qL)^\perp$ is constant on e -fibres.*
2. *For a specified family $\mathcal{T}_S$ of traces satisfying the original safety requirements, every available endpoint is in S if and only if $L(D \cap \mathcal{T}_S) \subseteq S$.*
3. *A claimed required endpoint additionally needs $\mathcal{T}_S \subseteq D$, or the specifically declared required subfamily in place of $\mathcal{T}_S$, and its original constructor warrant. These requirements are not consequences of finite-stage safety.*

Proof. Part (i) is theorem 3.3 with definedness retained. Part (ii) is the exact set form of the claimed endpoint property. Part (iii) spells out availability of every required application and the original warrant condition, neither of which occurs in the finite-stage induction. Thus the displayed criteria are necessary and sufficient for the stated information and membership claims, not a general existence proof for limits. $\square$

Proposition 7.4 (A sufficient topological route). *If X has a specified topology, S is closed, and a net entirely in S converges to an X -valued endpoint x , then $x \in S$. If q is continuous into a Hausdorff space, the net of profiles converges to $q(x)$ and has no other limit. These facts still do not establish that an endpoint exists for every proposed trace or that a proposed limit operation is licensed.*

Proof. If $x \notin S$, the open complement is a neighborhood of x and convergence would eventually place the net there, contradicting its membership in S . Continuity transports convergence. Distinct limits in a Hausdorff space have disjoint neighborhoods, contradicting eventual membership in both. $\square$

Closedness cannot be omitted. In $X = \mathbb{R}$, take $S = (0, \infty)$ and the warranted operation $f(x) = x/2$ on S . Every finite iterate of 1 is in S , but its real limit is $0 \notin S$. The example refutes a general limit-safety inference, not the legitimate finite operations. Likewise a program that

remains safe forever may never emit the promised result. If eventual emission is an original use, its truth must be retained as an original coordinate and proved by an appropriate termination or productivity argument [4].

7.3 Well-founded infinitary construction

There is a second genuine infinitary extension of finite assembly. Consider a set-sized rooted tree whose child relation is well-founded, so well-founded induction applies, while a node may have infinitely many children. Supply at each node a constructor of that arity, a compatible original domain, and a warranted output for every actual tuple of child results. For quotient evaluation, require exact descent on the attained tuple interface as in theorem 4.2; no surjectivity of an arbitrary infinite product is silently assumed.

Proposition 7.5 (Well-founded assembly). *Under these conditions, every existing evaluation from safe leaves has safe outputs at every node if each constructor preserves safety on its actual joint tuples. Two such evaluations with equal retained leaf values have equal availability and retained outputs if all node constructors satisfy exact descent. For the latter statement, use the specified strict semantics for unavailable children.*

Proof. Well-founded induction on nodes gives the asserted properties for every child before applying the local constructor hypothesis at the parent. The same argument applies at nodes of infinite arity because the premise quantifies over the entire actual child tuple. The statement is conditional on supplied interpretations and existing evaluations; it does not select a family of outputs or prove a solution to an equation with an infinite dependency branch. $\square$

Thus finite syntax, well-founded infinitary assembly, arbitrary-arity grounded inference, and convergent endpoint formation have distinct exact routes. A universal interaction theorem must specify which route covers the claimed construction; the word “iteration” is not that proof.

8 Witnessed exhaustion and the operator-closure theorem

The original question determines which obligations a proposal claims to satisfy. Exhaustion is obtained by negating their exact criteria, not by requiring every future technique to have a familiar name. We first state the defect theorem, then the closure result it supports.

Theorem 8.1 (Exhaustion of specified operator-realization claims). *Fix the independent data of Sections 2–7. For each of the following claims, failure has exactly the indicated kind of witness:*

1. Exact original result graph: *a surplus proposed record or an omitted required original record.*
2. An answer from retained information: *two enriched inputs with the same retained input but different required totalized answers. Missing source coverage is a separate defect witnessed by an uncovered original input.*
3. Exact partial operator or joint constructor descent: *an applicability collision or a retained-output collision at equal retained input profiles.*
4. Conservation of the original interior by an added repertoire: *a safe original state with failed local readiness or warrant, or a new edge followed by a finite original bad suffix.*
5. Conservative added rules at fixed grounds: *an added rule whose premises lie in the original closure and whose conclusion does not. For uniform conservativity the witness is a macro (A, j) with $j \notin \text{Cl}_{\mathcal{R}}(A)$.*

6. Erasure of a retained historical defect: *the originally identified failed occurrence that the proposed account must preserve but denies or drops.*

These alternatives are exhaustive for the listed claims and can overlap. An entirely different claim requires its own original semantics. An unestablished hypothesis is not automatically a witness that the claim is false.

Proof. Part (i) is theorem 2.3. Parts (ii) and (iii) negate the exact fibre and domain conditions of theorems 3.1, 3.3 and 4.2; failure of image coverage gives an input outside the image. Part (iv) is theorem 5.6, including the classical finite-path characterization. Part (v) is the negation of theorems 6.3 and 6.5; in the uniform case the macro's premises supply the witnessing ground set. Part (vi) uses the stipulated retention and proposition 4.6. Each proof identifies the original assertion contradicted by the witness. $\square$

For part (vi), dropping an occurrence is defective only when retaining it is required. An endpoint-only task need not reproduce an entire history. Likewise a surplus code defeats minimality, not necessarily completeness or soundness. The theorem does not confuse absence of a known witness with proof of universal correctness.

Theorem 8.2 (Closure by exhaustion for same-scope operator authority). *Let a non-degenerate determinate construction have an independently specified original answer frame, original result graphs, reuse contract, and sound grounds and rules for the relevant warrants. Fix an interpreted candidate operator universe $\mathcal{E}$ and a candidate rule universe M . Then:*

- (i) *Every sound complete realization of an original answer realizes the same graph. No purported same-answer totalization can add a record at an originally unavailable input; no loss of a required distinction can be repaired by sole-input postprocessing.*
- (ii) *The operators preserving the original greatest interior are exactly the subrepertoires of the unique greatest $\mathcal{E}_S$. Every simultaneous collection of compatible additions, and every finite mixed warranted execution with the stated readiness, preserves that interior.*
- (iii) *The rules preserving the original consequence closure in every ground environment are exactly the subfamilies of the unique greatest M_{unif} . They can be composed and added in arbitrary set-indexed families without supplying a new independent license.*
- (iv) *An unseeded license family all of whose entrances depend on a member of that family contributes no member to grounded closure. Any actual grounded entrance obeys theorem 6.7, even with infinite premise sets.*
- (v) *Exact partial actions and joint constructors preserve their attained interface semantics under composition; faithful representation changes transport the specified action. Required original-history coordinates cannot be removed by endpoint re-encoding. Limit results and infinitary assemblies obey the additional criteria of Section 7.*
- (vi) *A proposal violating any of these precise realization or conservation claims has the corresponding witness in theorem 8.1, or fails an explicitly additional coverage, existence, or limit obligation. A finite generated-language proof covers all original uses only through theorem 4.5; the semantic same-answer results themselves require no finite generation assumption.*

Consequently a purported unchanged realization cannot acquire a new authority to change the original answer, interior, or grounded consequence merely by adding an operator name, composing methods, moving through representations, mutually licensing rules, or deleting required history. Legitimate new construction is possible, but its original domain, answer, and warrant remain subject to these obligations.

Proof. Part (i) follows from theorems 2.3 and 3.3 and corollary 2.4. Parts (ii) and (iii) are theorems 5.5 and 6.5 and corollary 5.7; they are quantified over the independent candidate universes, not a list of technique names. Part (iv) is theorems 6.6 and 6.7. Part (v) combines theorems 4.1 to 4.3, 7.2 and 7.3 and propositions 4.6 and 7.5. Part (vi) applies the proved defect alternatives and the explicitly stated coverage bridge. Each transformation used in the closing conclusion falls under one of these semantic claims: if it purports to answer the same original question, exact graph realization applies; if it changes input, use, or grounds, those changes must be independently exposed rather than silently credited to the original claim. A method outside a proved generation scheme is still governed by the semantic tests, but is not asserted to belong to that scheme without its coverage proof. $\square$

8.1 Why the closure is substantive

Original-answer agreement alone is elementary: two exact realizations of one graph agree. The work of this paper is identifying when a proposed representation or operation can realize that graph, proving the precise composition conditions, constructing maximal conservative additions on both the state and authorization sides, and supplying witness forms that remain usable in a downstream application. The maximal repertoires are derived from independent local obligations and support arbitrary unions; neither is stipulated to be a closed list of familiar methods.

The claim is also not that every set-theoretic operator is realizable. The semantic universe can be broad, while a domain’s executable subset is separately constrained. On the other side, inability to compute an answer does not negate a determinate original incidence. The theorem addresses the claimed result and its warrant, whether established by explicit computation, a mathematical existence proof, or an appropriately justified physical realization.

8.2 How to test a claimed application

A test of a claimed application must specify the original problem, its incidence interpretation, and which asserted condition or conclusion it challenges. A failed hypothesis defeats that application, not the conditional theorem. A counterexample to a theorem itself must satisfy all its hypotheses and falsify its conclusion. A claimed exact descent is defeated by an equal-retained-input collision. A claimed conservative extension is defeated by a safe source with a local or successor defect. A claimed uniformly conservative macro is defeated by failure of derivability from its own premises. A new grounded entrance changes the no-entry hypothesis and can be legitimate; it does not contradict the no-entry theorem. An actual discrepancy between two independently warranted complete answers at the identical determinate incidence would challenge original-incidence conservation itself.

Relabeling a role or packaging several roles in one device does not evade their functional requirements. A new successful trial is also not a counterexample to retention of an old failure. Conversely, the author may not dismiss a demonstrated collision as “scope change” without identifying the changed original coordinate. These reciprocal obligations prevent both an easy escape from closure and an empty immunity to criticism.

9 Family exhaustions as applications of the universal tests

The family registry identifies each proposed source of additional authority and the conditions under which its claim succeeds or fails. Theorem 8.2 supplies the universal quantifier. The table and applications below identify the relevant test and the additional domain proof each proposed method must supply.

9.1 Application map

Proposal	Required proof	Defect if the stated condition fails
Selector or canonical representative	Eligible stabilizer-fixed output; correct original interpretation	Chosen output moved by an input stabilizer, or erased-parameter collision
Weighted or typical result	Original weighting data and exact erasure, if erasure is claimed	Same retained input, differing required result
Completion or evaluator	Exact original graph and covered domain	Surplus or omitted original record
Model or generic witness	Original existence meaning; selection or invariance when claimed	Missing warrant, coverage, or return factorization
Finite operator pipeline	Primitive and joint descent; readiness and sound warrants	Joint collision, illegal edge, or unsafe continuation
Infinite or stabilized output	Endpoint existence, domain, profile, and interior conditions	Missing endpoint or violated limit-constructor criterion
Repair or isolation	Same-incidence retention and required joint answers	Lost original failure or lost joint distinction
Meta-authority or derived rule	Independent sound grounds or conservative derivability	Ungrounded entrance or nonconservative macro witness

The table applies the established criteria; the following cases explain their domain interpretation. An unestablished application hypothesis remains an open obligation, and a failure label does not substitute for an actual witness.

9.2 Selection, canonical representatives, and weighting

A selection problem supplies an input, an eligible candidate set, and lawful transport of both. If no eligible candidate is fixed by the input stabilizer, an equivariant selector is impossible by Section 3. If the requisite fixed candidate and orbitwise data are supplied, selection can be legitimate. The question is then whether this is the original selection problem or one enlarged by an extra anchor.

Suppose a ranking or weighting parameter α is introduced and a selected result $r(x, \alpha)$ is returned as if it depended on x alone. The allowed parameter pairs form Z , and forgetting α is e . By theorem 3.3, a unique factorization is determined on the attained original inputs exactly when the totalized selected answer is constant on every e -fibre. Coverage of every claimed original input is a separate obligation. If constancy fails, two permissible choices give different totalized answers at the same retained input; if coverage fails, an original input has no represented source. Calling the parameter “canonical,” “typical,” or “minimal complexity” does not remove this witness.

An invariant weight can still have several minimizers. Its invariance identifies a minimizer set, not necessarily an element of that set. A further selection has the same stabilizer and fibre obligations. Conversely an originally warranted unique minimizer can supply a valid construction. Metrics and probabilities are not prohibited merely by their names; their original calibration, comparison class, domain, and relevance must be supplied. The pre-metric AMetric obstruction applies when such discriminatory work is claimed without licensed data [7].

9.3 Completion, total evaluators, compactness, and saturation

Let a be the original partial answer and let a completion method propose T . The graph exhaustion is exact: $T = G_a$, or there is a surplus or omission. In particular, an answer at $x \notin \text{dom } a$ cannot be an exact same-answer realization. If the method changes the codomain, supplies new input, or solves a different completion problem, its interpretation must be given. A map into a larger space is not illicit simply because its codomain is larger; re-entry as the old original answer requires the proved forgetting and graph conditions.

Completion can also compute a result already warranted by a complete specification but not previously computed. Such a construction need not be a redescription of its input. Its existence, domain, and output theorem provide the warrant. If it is offered for repeated reuse, theorem 5.4 tests its added obligations on the original interior.

Compactness and saturation arguments involve their own hypotheses about a family of constraints or types. A proved local-to-global theorem can be a sound original rule. If introduced as a reusable macro, theorem 6.5 tests derivability from its premises relative to the given calculus. If the calculus is incomplete, a new independently sound rule may extend its consequences without falsifying any original semantic answer. It is an addition of justified inferential resources, not a deduction of its own soundness from closure. The compactness hypotheses belong to the domain proof.

9.4 Genericity, forcing, and model witnesses

For a proposed forcing or model construction, identify the original base data, the enriched realizations, and the returned question. These determine a forgetting map $e : Z \rightarrow X$ and a totalized proposed answer r . If the answer varies among enriched realizations with the same original input, theorem 3.3 blocks its return as a uniquely determined sole-input answer. If it is invariant and its original meaning is warranted, it can return legitimately. A claim of invariance itself requires a proof over the relevant realizations, not merely a chosen example.

Existence, a named witness, and a canonical witness are different outputs. An existence theorem can be sound without selecting an individual witness. A selected witness needs the additional selection data; a canonical selection needs the appropriate transport property. If the application requires an executable witness, a semantic existence result alone does not provide that implementation. Conversely, the absence of an implementation does not refute the existence theorem.

An interpretation-and-return method is covered by theorem 7.2 when the relevant maps are faithful bijections, and by the exact descent criterion when they are quotients. A foreign evaluator cannot be credited with resolving old unavailable inputs merely because it assigns values in its own regime. The return interpretation must establish the same original domain and answer. This preserves the useful same-scope non-import implication without declaring every model witness inadmissible.

9.5 Infinity, iteration, and stabilization

There are at least four mathematically different proposals: a finite terminating computation, an infinite trace with qualified finite stages, a well-founded infinitary assembly, and a new endpoint obtained by a limit or fixed-point argument. Their proofs are respectively theorem 4.3 and corollary 5.7, the finite-stage consequence of corollary 5.7, proposition 7.5, and the explicit endpoint obligations of theorem 7.3. Grounded inference with infinitely many premises is separately covered by Section 6.

A stabilization rule must establish that the specified value stabilizes and that the selected value answers the original question. A convergence rule must establish its domain, endpoint, and any required uniqueness. Neither can acquire those facts by calling the trace already complete. At the same time, a properly proved limit is a possible warranted construction; the theorem's no-import conclusion targets unsupported authority, not infinite mathematical objects as a class.

9.6 Repair, temporal deferral, and subsystem isolation

Three repair claims must be separated. A corrected realization may now include a record already in the original graph. A changed object may qualify at a new incidence. A proposal may claim that an earlier failed incidence itself was successful. The first is tested by graph soundness and completeness; the second by its new domain and warrant; the third contradicts retained original history when its bearer and question are unchanged.

Temporal delay does not turn a surplus original record into a correct one, but later evidence can establish an answer that was previously unknown. Physical irreversibility of a retained occurrence does not require an observer to know its result at its formation. A deferred proof and a deferred qualification are different claims.

Subsystem isolation is assessed by the original task. If the requirement concerned the whole joint construction, projecting to a successful component cannot establish that whole requirement; a fibre collision can expose the lost joint information. If the original question genuinely concerns only that component and factors through the projection, the projection is legitimate. Thus the theorem blocks evasion of a required obligation without banning every modular repair.

9.7 Foundation changes and meta-level authorization

Moving to a new language does not by itself add an original answer. A faithful translation preserves that answer by proposition 7.1. A transported action retains its specified semantics by theorem 7.2. Any genuinely new premise or axiom must expose its independent warrant and its relation to the original task.

A proposed rule which authorizes the evaluator solely because the evaluator endorses that rule is blocked by theorem 6.6 when the license family has no grounded entrance. The result also covers multiple mutually supporting authorities and infinitely many such rules. If independent semantics establishes the new rule's soundness, that evidence must be presented as the ground it is; the theorem does not relabel a legitimate soundness proof as an illicit import.

10 Worked tests and downstream exports

10.1 Bridge records: four failures and one legitimate construction

Let $X = \{s, t, f, r\}$ represent a robust bridge, a fragile bridge, an original failed configuration, and a repaired bridge in a new trial. Fix local qualification $A = \{s, t, r\}$ and original warranted transitions $s \rightarrow s$, $t \rightarrow f$, and $r \rightarrow r$. Then $B = A$ and

$$S = \{s, r\}.$$

The fragile configuration passes the present local test but fails the declared repeat-use contract; its path $t \rightarrow f$ witnesses its exclusion. The labels are schematic original states, not an assertion that a local bit fully describes a physical bridge.

1. A claimed complete robustness graph reporting $(t, 1)$ has a surplus result. The actual answer for the declared reuse question is 0, witnessed by $t \rightarrow f$.
2. A proposed complete interior $\{s\}$ omits r . The singleton $\{r\}$ is an independently sound region, witnessing the omission. As a partial sound region, $\{s\}$ remains legitimate.
3. Retaining only current local qualification loses the distinction between s and t . The retained codes agree, but their robust-use answers differ. No sole-input decoder from that bit realizes both original answers.
4. An added candidate with edge $s \rightarrow t$ and otherwise satisfied local conditions is outside $\mathcal{E}_S$: the new edge followed by $t \rightarrow f$ is its finite unsafe witness. This remains so even though its immediate target passes the local test.
5. A candidate swapping s and r , with independently established warrants, preserves S and may belong to $\mathcal{E}_S$. It is a nonidentity transformation of states. It does not identify the old failed trial at f with the successful repaired trial at r ; a history preserving both occurrences keeps their different results.

Adding a finger support changes the tested conditions. If the original question was unsupported robustness, the support cannot establish it without a theorem showing that the support is irrelevant to that very question. The mathematical test is an original graph or fibre test, not a linguistic judgment about whether the intervention sounds small.

10.2 Why individually safe systems can still interfere

Take $X = \{a, b, z\}$ and local validity $B = \{a, b\}$. One repertoire has only $a \rightarrow b$, and another only $b \rightarrow z$, with all edges warranted. Their separate interiors are $\{a, b\}$ and $\{a\}$, respectively. The joint interior is empty, because the mixed path $a \rightarrow b \rightarrow z$ fails. This does not contradict theorem 5.5: the two proposals were not both checked as preserving one common original interior. When every candidate is tested on the same S , that theorem covers all mixed finite paths.

Shared resources, mutable context, and history-sensitive licenses must be represented in the original states and candidate obligations if the intended use depends on them. Adding another operation must not silently change the meaning of a package already tested. Otherwise the proposed state model omitted a required distinction, and the local hypotheses of the simultaneous-extension theorem have not been established.

10.3 A small typed operator language

Let the Boolean carrier be $X = \{0, 1\}$ with original value observation $q(x) = x$. Include total negation $n(x) = 1 - x$ and a partial assertion $p(1) = 1$ with domain $\{1\}$. All their defined applications can have their original warrants. The language is nontrivially partial because $p(0)$ is unavailable.

With this complete q , both operations descend; negation acts nontrivially on Q . The composite $p \circ n$ is available only at 0, whereas $n \circ p$ is available only at 1. Replacing q by the constant “well-formed Boolean” code makes the domain of p nonsaturated. The equal-code inputs 0, 1 witness the lost applicability distinction. A total evaluator which returns a Boolean as the same original $p(0)$ would add a surplus record; an evaluator returning a tagged unavailable result merely represents the original partiality explicitly.

The two-negation program and identity have the same endpoint function. If the original task records intermediate states or costs, their complete contextual profiles can differ. Thus extensional

endpoint equality does not erase original execution obligations. Conversely their equal endpoint behavior is a genuine equivalence when that is the complete original use family.

10.4 Exact bridges to uniqueness and algebraic rank

Downstream applications sometimes need more than no hidden authority. The following elementary bridges state the extra work exactly.

Proposition 10.1 (Unique occupant from a separating readout). *Let Z be compatible realizations, $\iota : Z \rightarrow O$ their original occupant identities, and $v : O \rightarrow Y$ injective on $\iota(Z)$. Then $\iota(Z)$ is a singleton if and only if $Z \neq \emptyset$ and $v\iota$ is constant on Z .*

Proof. A singleton identity image gives nonemptiness and constant readout. Conversely constancy and injectivity identify any two attained identities; nonemptiness supplies one. $\square$

Proposition 10.2 (The cyclic-generation bridge for a rank-one claim). *Let V be the vector space over a field k of original primitive consequences, with its algebraic interpretation independently established. If a proved linear map $\lambda : k \rightarrow V$ is surjective, then $\dim_k V \leq 1$. If also $\lambda(1) \neq 0$, then $\dim_k V = 1$. Equivalently, a nonzero $\omega \in V$ gives dimension one exactly when every original consequence lies in $k\omega$. More generally, a surjective linear map $k^m \rightarrow V$ gives $\dim_k V \leq m$. An independently faithful linear isomorphism preserves these conclusions under redescription.*

Proof. Linearity gives $\lambda(a) = a\lambda(1)$, so surjectivity makes V the span of one vector. A nonzero spanning vector is a basis. In the general case the images of the m standard basis vectors span V . Linear isomorphisms preserve spanning and linear independence by applying the map and its inverse. $\square$

For modules, a corresponding surjective homomorphism from a cyclic module establishes cyclic generation; a particular numerical rank requires the ring and rank convention appropriate to the application. Functional factorization through a scalar is not enough. On $\mathbb{R}$, the functions $x \mapsto x$ and $x \mapsto x^2$ both factor through x , but are linearly independent as real-valued functions: evaluating $ax + bx^2 = 0$ at 1 and -1 gives $a = b = 0$.

Thus corollary 3.4 can establish that no additional original information is missing once completeness of a proposed residue has been proved. Proposition 10.2 states the separate algebra-compatible bridge needed for rank one. Operator closure then protects these independently established structures against unauthorized additions or presentation changes. A dimension, a numerical constant, or a unique physical occupant requires the corresponding domain proof.

10.5 Export map

The following table states invocable results and the application facts needed to use them. Theorem 8.2 is the synthesis; theorems 5.5 and 5.6 supply repertoire saturation and its defects, while theorems 6.3 and 6.5 to 6.7 distinguish fixed-ground conservation, uniform reuse, and grounded authority. For a complete original answer frame, these semantic results apply to every presentation realizing that frame. Applying them to a generated-language implementation requires a proof of its coverage of the original task.

Result	Required domain proof	Consequence preserved
Theorem 2.3 and corollary 2.4: exact graph	Original input identity, answer meaning, and actual partial domain; claimed coverage	No surplus old-scope result; no repair of structural unavailability by re-encoding
Theorem 3.3 and corollary 3.4: information closure	Full relevant forgetting fibres, original answer warrant, and input coverage	No independent returned answer from distinctions that were erased; no hidden extra coordinate after proved completeness
Theorems 4.2, 4.3 and 4.5: assembly	Joint applicability, retained-output congruence, and interpretation of all claimed original uses	Closed finite composition and contextual reuse; no inference of joint compatibility from marginal bits
Theorems 5.5 and 5.6: maximal repertoire	Independent original A, R, W , candidate meanings and coverage, readiness and warrants on the same S	Unique greatest compatible repertoire, arbitrary simultaneous additions, exact defect witnesses
Theorem 6.3 and corollary 6.4: fixed-ground conservation	Original closure, added premises, and interpreted rules	Exact conservation iff added grounds lie in the old closure and all added rules preserve it
Theorem 6.5: derived-rule saturation	Original rule meanings; each macro derivable from its own premises	Uniform preservation of original consequence closure, including arbitrary premise sets
Theorems 6.6 and 6.7: no bootstrap	Independent grounds and actual entering-rule dependencies	No ungrounded mutual license; any grounded family has an actual grounded entrance
Proposition 4.6: retained incidence	Original occurrence identity and a proved retention map	An earlier failed trial is not made successful by later repair or compression
Theorem 7.2: transport	Faithful interpretations and required structure, domain and warrant transport	Same action and corresponding fixed points/interiors; existence hypotheses remain independent
Theorem 7.3 and proposition 7.5: infinitary construction	Endpoint existence and domain, limit compatibility or well-founded assembly conditions	Exact extension beyond finite syntax where those hypotheses are proved
Propositions 10.1 and 10.2: stronger endpoints	Nonempty realization and separating readout; or surjective linear map $k^m \rightarrow V$ over a specified field	Occupant uniqueness or rank bound with the domain bridge explicitly supplied

10.6 Corpus application patterns

The intended uses include ruling out a hidden charge operator that purports to answer an unchanged charge question, preventing numerical readiness from being promoted to a forced value without its witness, preserving a certified operator envelope, and blocking transport without an original channel warrant. The applicable theorem is selected by the actual claim.

For a covariant-derivative classification, the domain may supply a fixed first-jet carrier, normalization, linearity, and locality. Closure then protects the complete classification only after the domain has established those hypotheses and exhausted its operator possibilities. For a horizon or irreversible exchange claim, structural absence of a required certificate or warranted path can give a genuine obstruction. Lack of currently known evidence is not that nonexistence proof.

For a generation-cardinality or primitive-register claim, the no-extra-answer theorem is useful only after the proposed retained residue has been shown complete for all original primitive consequences. If the conclusion is algebraic rank one, the linear or module generation proof is

additionally required as in proposition 10.2. The presence of one complete kernel of qualification is not a count of the independent quantities a domain can possess. Each downstream conclusion thus follows from the closure theorem together with its stated domain bridge.

10.7 Conclusion

The closure theorem is an exhaustion of claimed original authority. Exact graphs determine which results can belong to the unchanged problem; complete interfaces determine which distinctions can be forgotten; the greatest conservative repertoire determines which additional obligations preserve the original safe interior; and grounded macro saturation determines which derived rules can be introduced without enlarging the original consequence relation. The associated witness theorems make each failure inspectable.

The theorem admits warranted construction of new calculations, proofs, safe operations, repaired objects, and correctly justified limits. Their domain and warrant cannot be supplied by reclassifying an old failure, silently changing the input, or selecting an ungrounded fixed point of mutual authorization. The Kernel's necessity governs every such determinate construction, while the present theorem specifies the operator-level consequences and their exact conditions of export.

A Sharpness and proof-dependency ledger

A.1 Small models isolating the hypotheses

1. *Standing is not identity.* On $X = \{0, 1\}$ let both states qualify. Negation preserves qualification and changes both states. A partial primitive $p(0) = 1$ supplies a proper operator domain if one is required for nondegeneracy. No auxiliary datum is needed. This shows why qualification or determined output cannot replace complete-profile preservation.
2. *Outputs alone miss partiality.* Let $e(x, \alpha) = x$ for $\alpha \in \{0, 1\}$, and let an operation return the same constant when $\alpha = 0$ but be unavailable when $\alpha = 1$. Its defined outputs agree, yet its totalized answer does not descend. Domain saturation is necessary.
3. *Output erasure does not authorize a trace.* Let both parameter choices produce a constant endpoint, but only one choice have the original action warrant. Endpoint factorization holds. A profile recording execution legality still distinguishes the choices.
4. *Joint compatibility is not marginal qualification.* Admit the Boolean pairs $(0, 0)$ and $(1, 1)$ only. Constant component codes lose the original joint domain even though each component is separately qualified.
5. *Empty interior is possible.* With $X = \{a, b\}$, $B = \{a\}$, and edge $a \rightarrow b$, the greatest interior is empty. Maximal repertoire existence therefore does not prove an admissible occupant exists.
6. *A dormant rule is not uniformly conservative.* With no original rules, grounds $\emptyset$, and macro $p \Rightarrow q$ for distinct judgments, the base closure is unchanged. At grounds $\{p\}$ the macro adds q , so uniform conservativity fails.
7. *Closed consistency is not grounded entry.* The rule $q \Rightarrow q$ admits the closed set $\{q\}$ but does not derive q from empty grounds. This is the smallest no-bootstrap example.
8. *A limit can leave the interior.* Iterating $x \mapsto x/2$ from 1 stays positive at every finite step and converges to 0. The endpoint requires the distinct limit-constructor test.

These examples show actual failure of a stronger conclusion when a needed hypothesis is removed. They do not prove that every hypothesis is logically independent of every other one in all possible applications. In particular, a particular domain may supply stronger assumptions from which several displayed conditions follow together.

A.2 Load-bearing dependency order

Result	Direct mathematical input	Subsequent use
Theorem 2.3	Independently specified original graph; set equality	No totalization; original-answer conservation
Theorems 3.1 and 3.3	Fibre constancy, including availability	Composition, selection return, foreign-interface tests
Theorems 4.3 and 4.5	Exact joint constructors; actual use interpretations	Finite assembly and original-use coverage
Lemma 5.2	Fixed A, R, W and universal safety meaning	Conservative extension and finite bad suffixes
Theorems 5.5 and 5.6	Greatest interior plus fixed candidate packages	Saturated repertoire and witnessed operator exhaustion
Theorems 6.5 and 6.7	Least rule-closed supersets; independent grounds	Uniform macro saturation and no circular authority
Theorems 7.2 and 7.3	Faithful maps; original endpoint constructor	Representation and limit exports
Theorem 8.2	All preceding exact criteria	Family applications and downstream export map

The family discussion, examples, and downstream bridges are not premises of the central closure theorem. Classical existence results are not inferred by assuming the completeness of an interface intended to prove them. The candidate sets and original interpretations must be supplied before the saturation tests are applied. These constraints make the claimed non-circularity inspectable at each proof step.

A.3 Verification boundary

The accompanying project includes finite-model checks of partial descent, composition, conservative repertoire extension, grounded rule extension, and the displayed small countermodels. These are checks of the formal statements in finite instances. The proofs in the manuscript establish the arbitrary-set results; finite computation does not replace their quantifiers, prove an application's original-task coverage, or validate a physical model. The review and dependency records accompanying the project state the source versions and any application obligations separately from the mathematical conclusions.

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